

Sophia Zhang

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EDUCATION

Massachusetts Institute of Technology <i>B.S. in Computer Science and Mathematics</i>	Cambridge, MA <i>Aug. 2025 – May 2028 (expected)</i>
Coursework: Inference & Information (G), Probability (G), Machine Learning (G), TinyML and Efficient AI Computing (G), Operating Systems, Computation Structures	
Princeton International School of Mathematics and Science (PRISMS) <i>High school</i>	Princeton, NJ <i>Sep. 2022 – June 2025</i>

EXPERIENCE

Undergraduate Researcher (UROP) <i>Dina Katabi's Lab, MIT CSAIL</i>	Sep. 2025 – Present Cambridge, MA
Working with diffusion models, multimodal foundation models, and recommendation models.	
AI Engineering Intern <i>Alpha School</i>	June – Aug. 2026 Austin, TX
Built RL (GRPO) training setups and evaluations for LLM tutors and educational apps, grounded in learning science and pedagogy.	
Trading & Technology Intern <i>Hudson River Trading</i>	Jan. 2026 New York, NY
Software engineering and quant research projects, including trading Brazilian equities.	
Research Collaborator – LLMs for Formal Theorem Proving <i>Microsoft Research Asia</i>	June 2025 – Jan. 2026 Remote
Verified Lean formalizations of math proofs and contributed to system improvements; two papers below.	

PUBLICATIONS

<u>STAR-PólyaMath: Multi-Agent Reasoning under Persistent Meta-Strategic Supervision</u>	NeurIPS 2026 (Spotlight)
Jiaao Wu, Xian Zhang, Hanzhang Liu, Sophia Zhang, Fan Yang, Yinpeng Dong	
<u>Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean</u>	ICML 2026
Linbin Tang, Jingyan You, Zilin Kang, Hanzhang Liu, Sophia Zhang, Zenan Li, Chenrui Cao, Liangcheng Song, Jiaao Wu, Xian Zhang, Fan Yang	

PROJECTS

AI Math Olympiad Progress Prize 3 <i>Python, PyTorch, vLLM, Unsloth</i>	Jan. – Apr. 2026
- Competed with a team of MIT freshmen, all Olympiad medalists. Built for the training and inference scaffolding (vLLM, Unsloth), hidden-state extraction, and parameter sweeps. - Authored the team's <u>technical writeup</u>.	

AWARDS

Putnam Top 200, 2026
USAMO Silver, 2025
USAPhO Honorable Mention, 2025
Mathematical Olympiad Program (MOP) Two-time attendee, 2023, 2024

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, Java, TypeScript, Lean, RISC-V Assembly, Minispec, Dafny
AI Research: RL post-training, vLLM, speculative decoding, diffusion models, multimodal foundation models
Tools: PyTorch, Unsloth, NumPy, SciPy, Pandas, Matplotlib, Jupyter, automated eval pipelines